

EgoThumb: A Vision-Driven Supernumerary Robotic Thumb for Autonomous Grasp and Holding Assistance

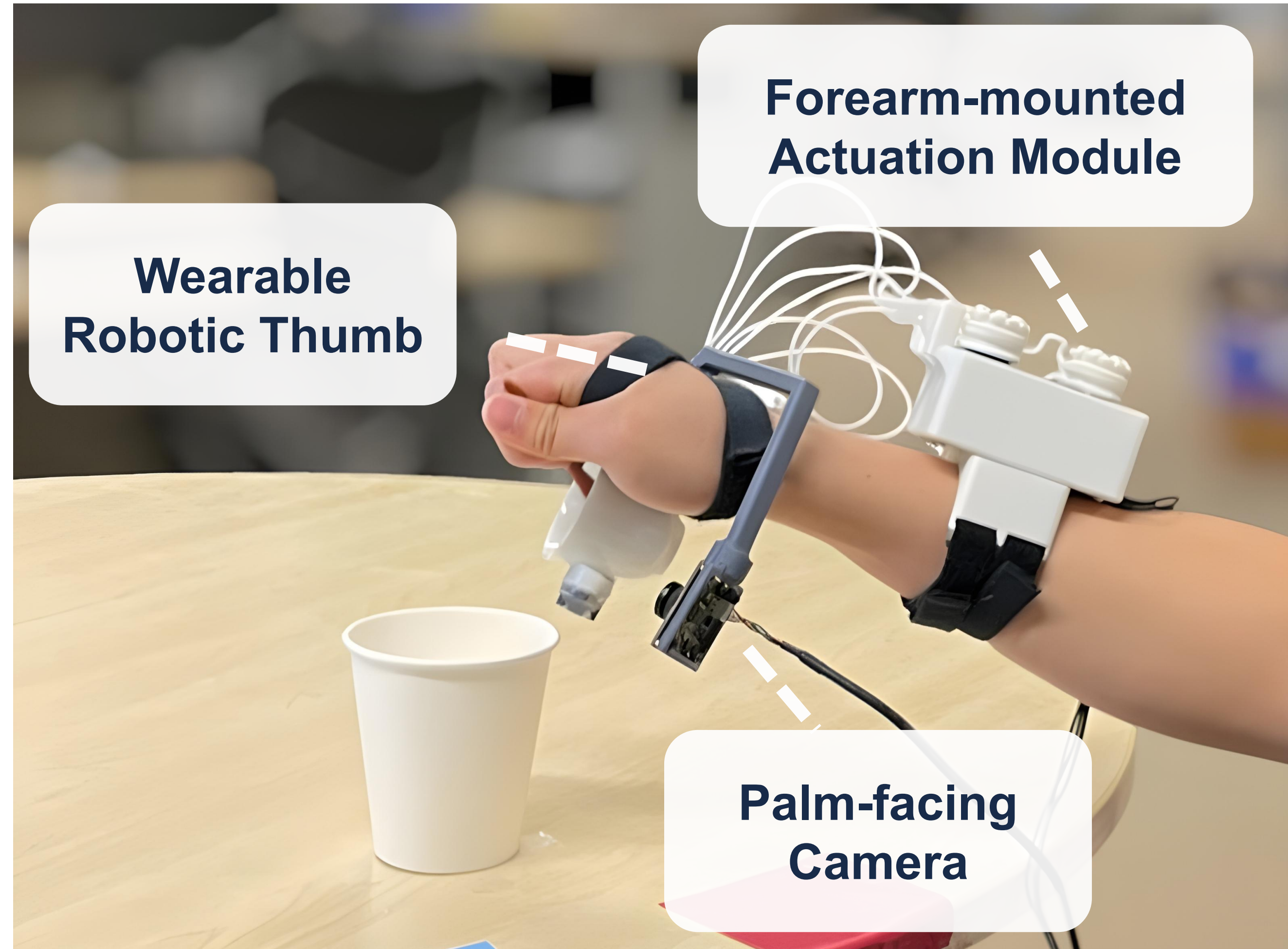
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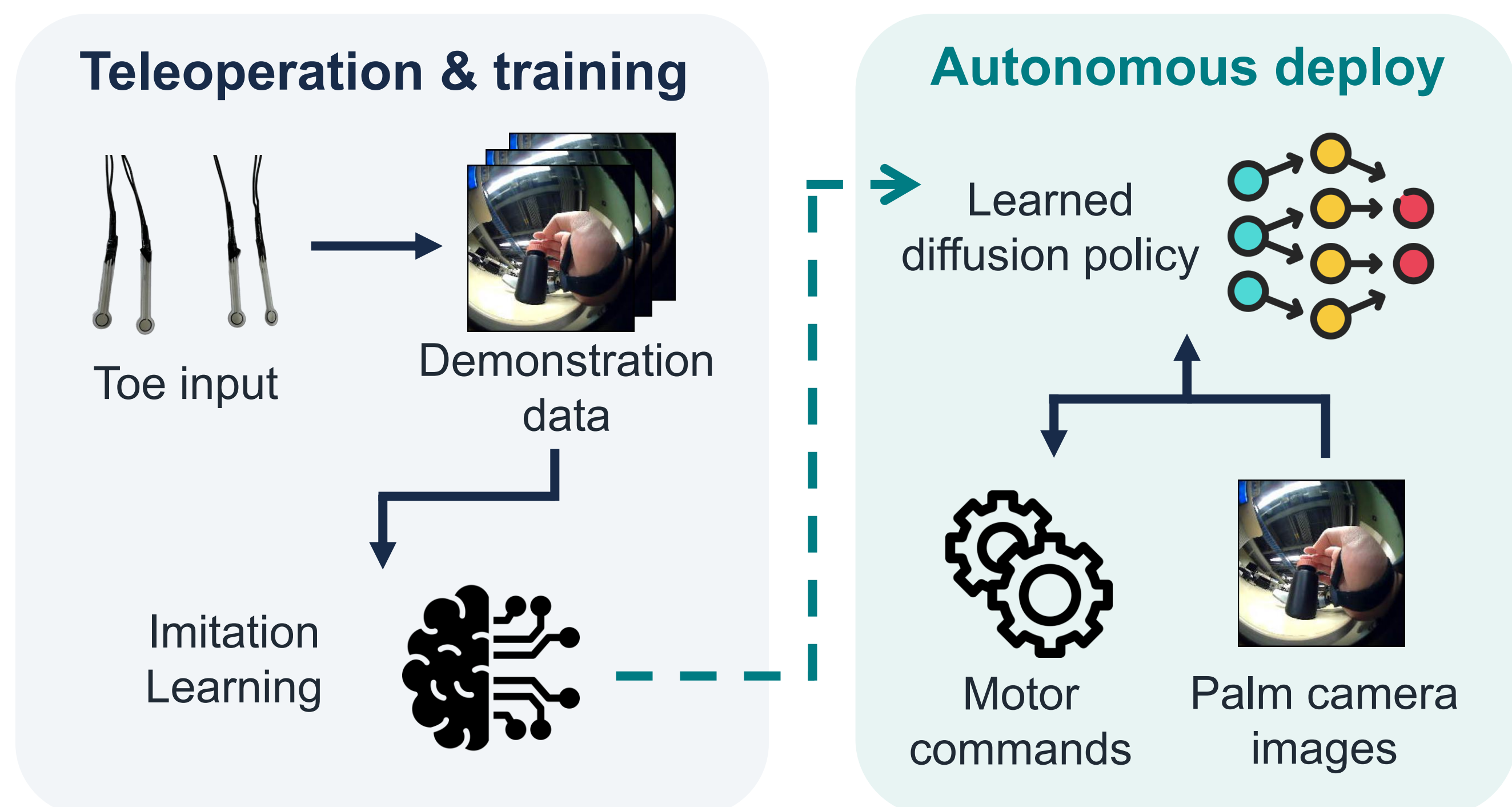
 ego-thumb.github.io

1 EgoThumb: System Overview

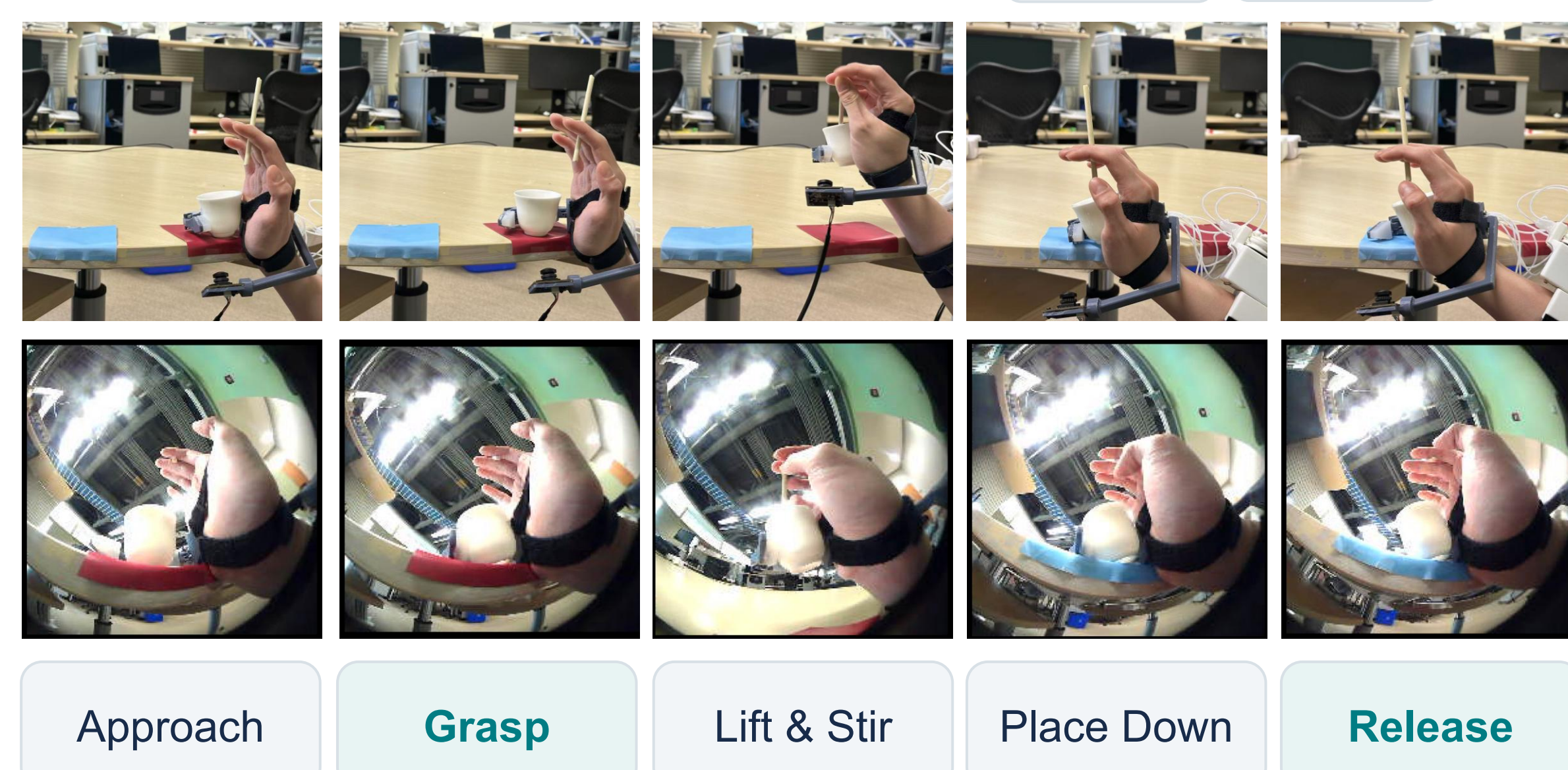
A wearable robotic thumb that works with you



How it works

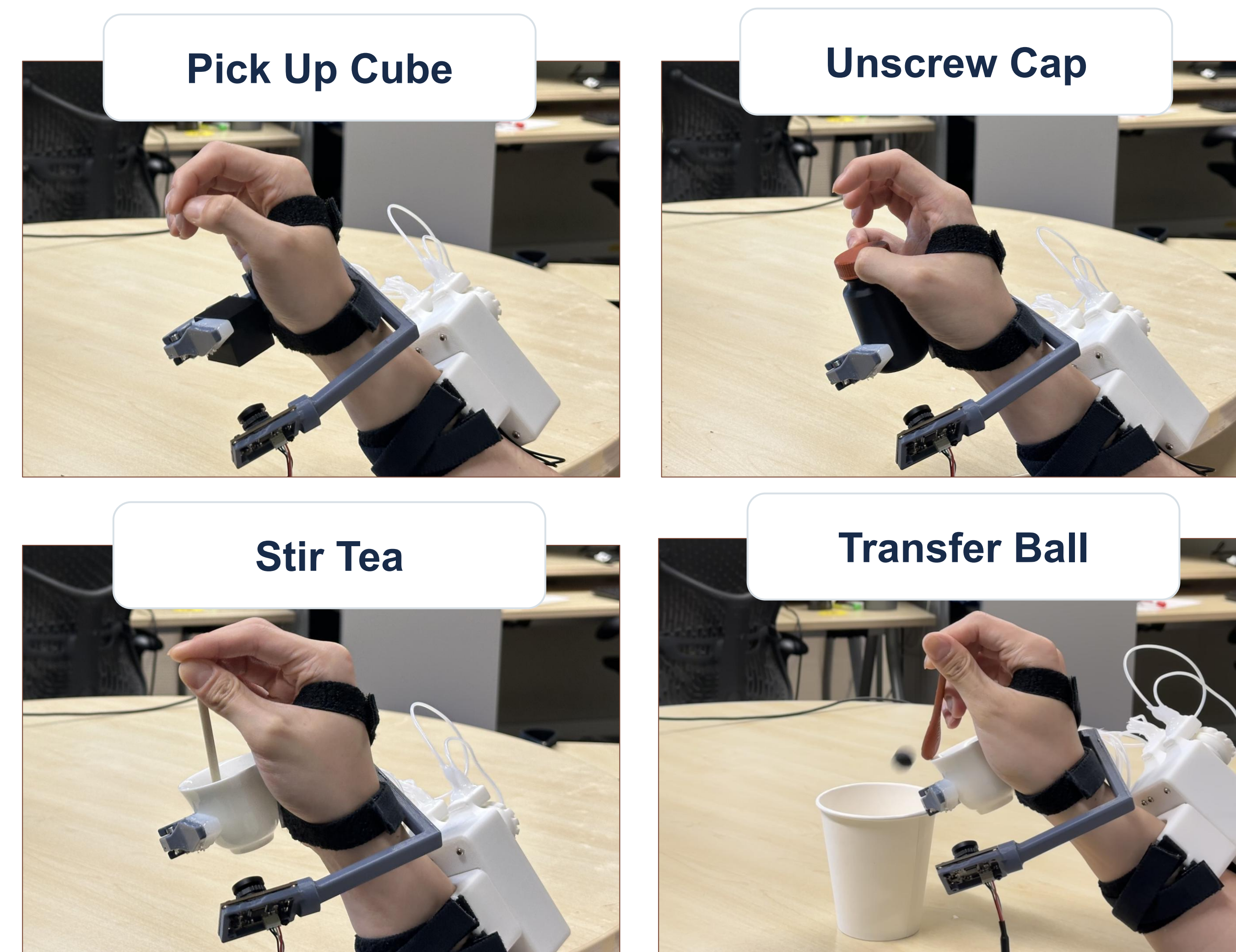


Example Interaction Sequence

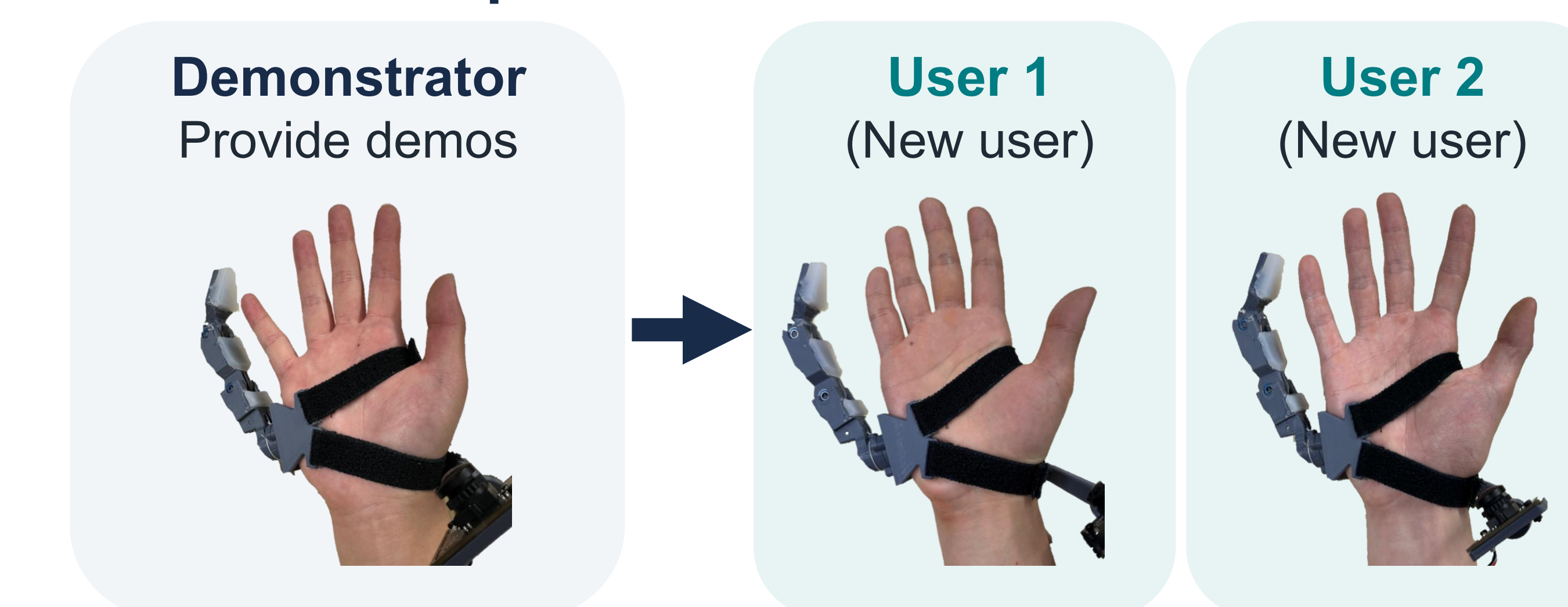


2 Generalization Across Users

Four “hold-and-manipulate” tasks



Evaluation setup



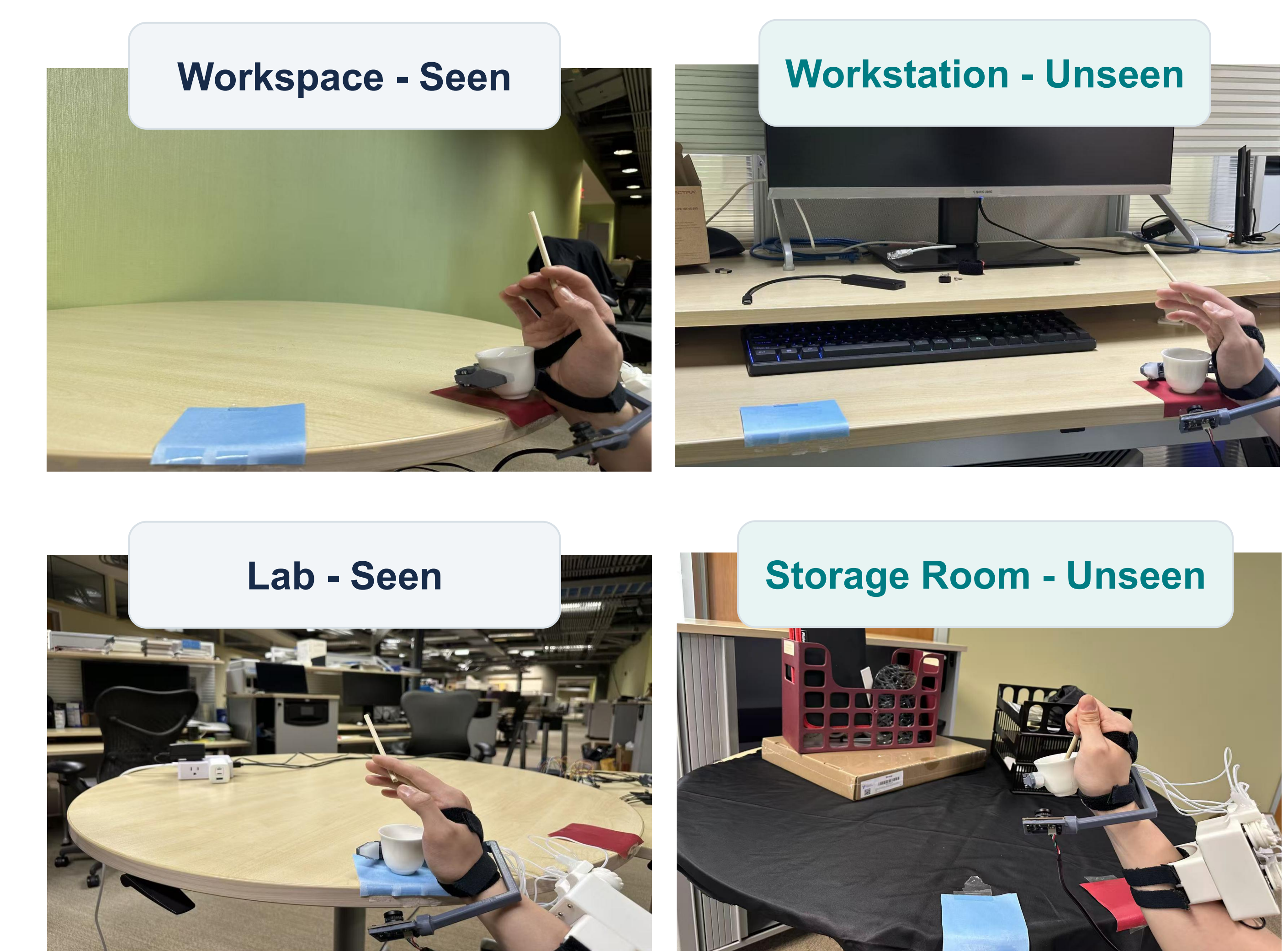
Task success rates (10 trials per task per user)

Task	Demonstrator	User 1	User 2
Pick Up Cube	100%	100%	100%
Unscrew Cap	70%	90%	80%
Stir Tea	90%	80%	80%
Transfer Ball	100%	100%	90%
Average	90%	92.5%	87.5%

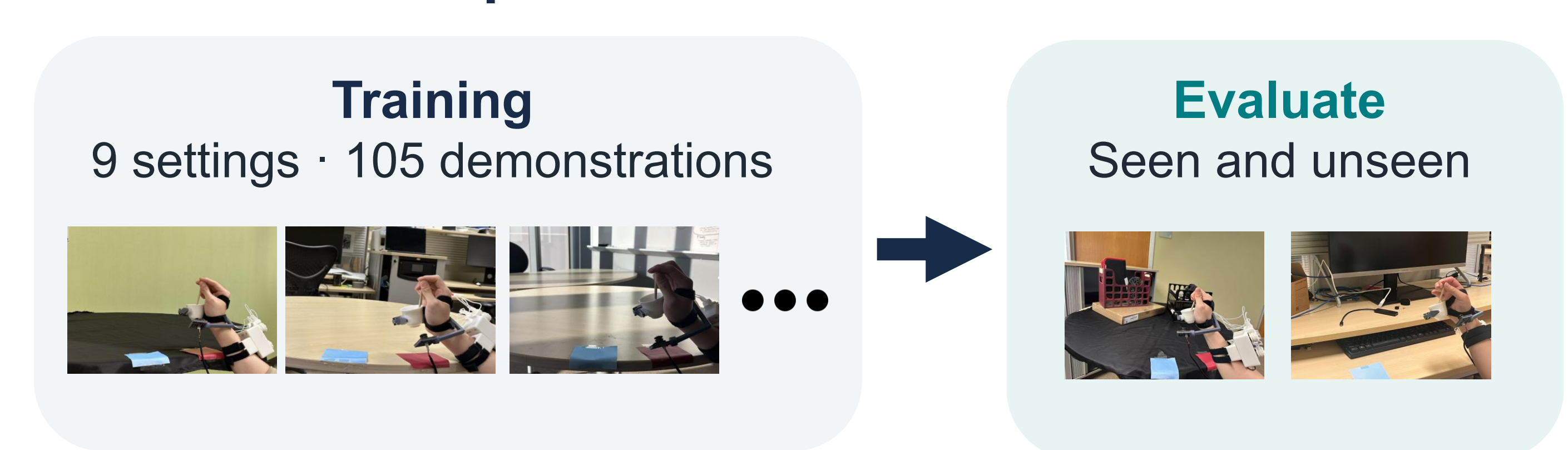
Strong performance across four tasks and three users

3 Generalization Across Backgrounds

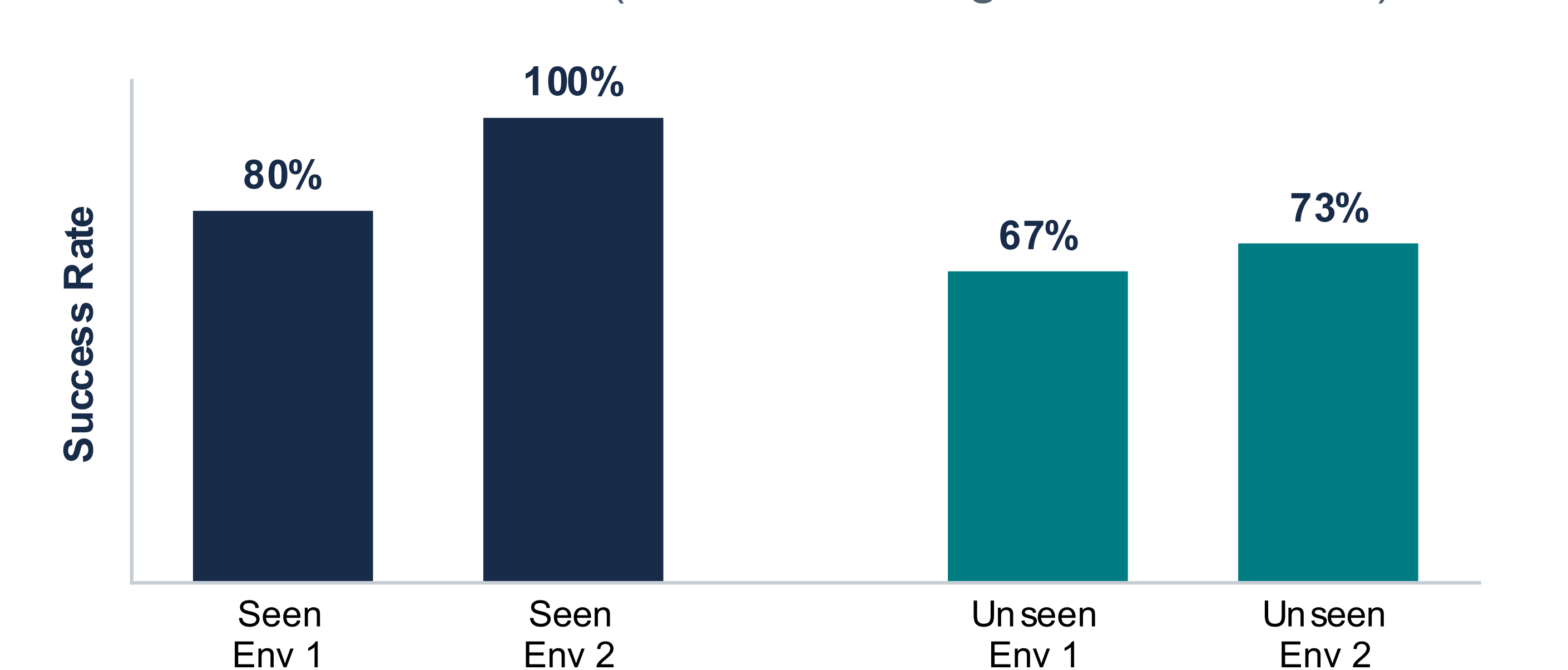
Robust to diverse environments



Evaluation setup



Task success rates (Stir Tea, averaged across users)



Successful manipulation across seen and unseen backgrounds